SEMINAR SUMMARY: JANUARY 20, 2026 - SEAN COX CAN WE PREDICT THE FUTURE?

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So we began with a bit of an odd question, namely "how robustly can you predict the future?" And clearly this is a problem that has long been sought after, take for example the dynamical systems of weather prediction. But I digress because the motivation here was to look at the results surrounding this question with a logical lens.

So our setting sets up as suppose we have a function that is unknown $F : \mathbb{R} \to \mathbb{R}$, but we know $F|_{(-\infty,t)}$, can we predict $f(t)$?

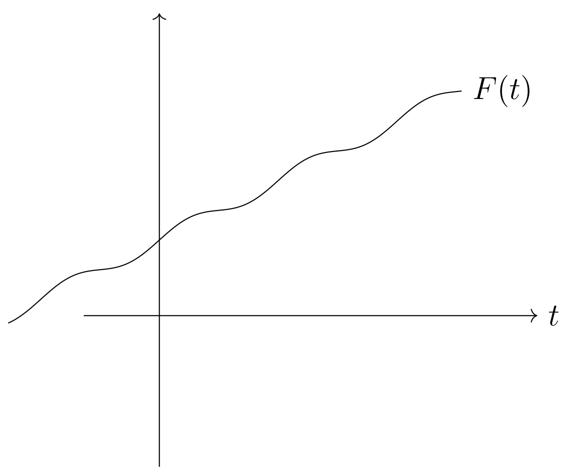

FIGURE 1. An arbitrary F with known quantity from the left

well this is a classical problem if the function is continuous, and we have that

$$F(t) = \lim_{s \nearrow t} f'(s) = L$$

But what do we do if F is not continuous. Or even worse consider $t = 0$ and the plot of $\sin(1/x)$ for $x < 0$

So we know that for *discontinuous* functions it is impossible to *always* predict$F(t)$ based only on the information given by the behaviour of F before t. However:

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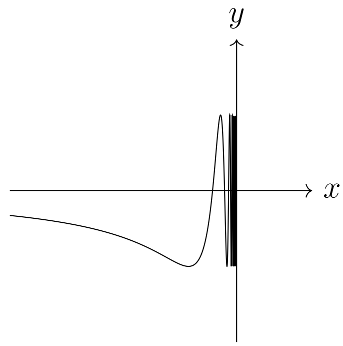

FIGURE 2. $\sin(1/x)$

Theorem. (Hardin - Taylor 2008) There exists some function $\mathcal{P}$ taking partical functions as inputs so that $F : \mathbb{R} \to \mathbb{R}$ (possibly nowhere continuous):

$$\mathcal{P}\left(F\big|_{(-\infty,t)}\right) = F(t)$$

for "almost every" t.

Then we were presented with the following question: Can we also arrange that $\mathcal{P}$ is "shift-invariant" with regards to time?

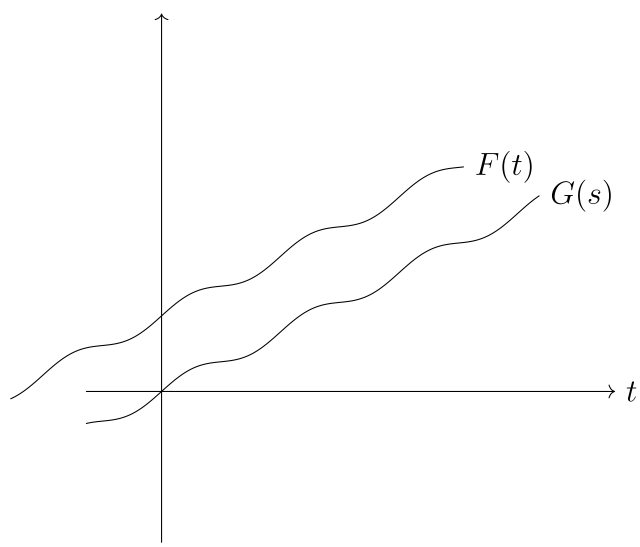

FIGURE 3. A shift of $F(t)$ by a constant to a function $G(s)$

Can we also arrange that:

$$\mathcal{P}\left(F\big|_{(-\infty,t)}\right) = \mathcal{P}\left(G\big|_{(-\infty,s)}\right)?$$

Theorem. (Bajpai - Vellemm 2016) Yes. Infact we can arrange $\mathcal{P}$ to be invariant with regards to distortions of time with assuming we have positive slope.

More precisely, we arrange that $\mathcal{P}$ has the property

$$\mathcal{P}\left(F\big|_{(\infty,t)}\right) = \mathcal{P}\left(F\dot{A}\big|_{(-\infty,A^*(t))}\right)$$

for any $A : \mathbb{R} \to \mathbb{R}$ where $A(x) = mx + b$ for some $m, b \in \mathbb{R}$ with $m > 0$. So what is the domain of $\mathcal{P}$? $\mathrm{dom}(\mathcal{P}) = \{f|$ for some $t \in \mathbb{R}$ with $\mathrm{dom}(f) = (-\infty, t)$ and $\mathrm{range}(f) \subseteq \mathbb{R}\}$

Theorem. There is no predictor that is invariant with respect to all increasing C^∞ distortions of time.

Note. There are C^∞ non-analytic function that work here. Specifically the "bump functions" which are functions that are continuous yet 0 except for on an interval. And it's curious that these get used in machine learning a good deal these days.

Theorem. (Cody-Lee-Cox) It's consistent with ZFC that there are *no* analytic-invariant prediction functions. (Infact, the continuum hypothesis implies no such predictors).

We then went a little into the Hardin-Taylor Proof (rough sketch):

Proof. We use the Axiom of Choice to fix a well order "$<$" of $\mathbb{R}^\mathbb{R} = \{$All functions from $\mathbb{R} \to \mathbb{R}\}$, i.e. $<$ is a linear order and has no infinite descending chain.

$$\mathcal{P}(f) = F_f(t_f)$$

where F_f is the $<$-least function extending f. We also see that $f : (-\infty, t_f) \to \mathbb{R}$. □